Amrita School of Artificial Intelligence, Faridabad
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Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 1.9

DMD and need of Hankelization

BTech 4th Sem

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*All Sources: Stanford University, Texas Tech University, UC Santa Barbara, etc.

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We observe a system over time:

$$x_1, x_2, x_3, \dots, x_m$$

Each x_k is the system state.

🎯 **Goal of DMD**

We want to find a matrix A such that:

$$x_{k+1} = Ax_k$$

👉 This means:

- system evolution is **linear**
- matrix A captures dynamics

📌 **Construct matrices**

$$X = [x_1 \ x_2 \ x_3 \ \dots \ x_{m-1}]$$

$$X' = [x_2 \ x_3 \ x_4 \ \dots \ x_m]$$

DMD Assumption

We assume

$$X' = AX$$

This means the matrix A maps each snapshot to the next snapshot.

📌 **Compute**

$$A = X'X^{\dagger}$$

👉 This gives **best linear approximation**

$$X^{\dagger} = \left((X^T X)^{-1} X^T\right)$$

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Higher-Order System

Example of Higher-Order System

Consider a system

$$x_{k+1} = 2x_k - x_{k-1}$$

This system depends on **two previous values**.

Standard DMD cannot capture this properly.

Tricks

"We trick the system by increasing dimension."

Instead of: x_k

We use: $\tilde{x}_k = \begin{bmatrix} x_k \\ x_{k-1} \\ x_{k-2} \end{bmatrix}$

Now system becomes: $\tilde{x}_{k+1} = A\tilde{x}_k$

Now DMD works!

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CASE: If the relation between state vectors are predefined

here: $x_{k+1} = 2x_k - x_{k-1}$

we define a **new state vector**:

$$\tilde{x}_k = \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$$

We define k to k+1:

$$\tilde{x}_{k+1} = \begin{bmatrix} x_{k+1} \\ x_k \end{bmatrix}$$

$$\Rightarrow \tilde{x}_{k+1} = \begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix}$$

So, We want a matrix A such that:

$$\begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix} = A \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$$

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We want a matrix A such that:

$$\begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix} = A \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$$

Assume:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then:

$$A\tilde{x}_k = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix} = \begin{bmatrix} ax_k + bx_{k-1} \\ cx_k + dx_{k-1} \end{bmatrix}$$

Compare with:

$$\begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix} = \begin{bmatrix} ax_k + bx_{k-1} \\ cx_k + dx_{k-1} \end{bmatrix}$$

First row:

$$ax_k + bx_{k-1} = 2x_k - x_{k-1}$$

So:

$$a = 2, b = -1$$

Second row:

$$cx_k + dx_{k-1} = x_k$$

So:

$$c = 1, d = 0$$

FINAL MATRIX

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

FINAL RESULT

$$\tilde{x}_{k+1} = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \tilde{x}_k$$

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IMPORTANT QUESTION

👉 “If we already found A, why do we need Hankel?”

Because:

👉 In real problems:

- we **DO NOT know the equation**
- we only have **data**

❗ **Here we manually derived A**

But in real DMD:

👉 ***we must compute A from data***

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START FROM ORIGINAL DATA (VERY IMPORTANT)

We are given a **time series**:

$$x_1, x_2, x_3, x_4, x_5$$

Example:

$$x = [1, 2, 3, 4, 5]$$

👉 Each x_k is a **measurement at time k**

A **state vector must contain ALL information needed to compute the next step:** $\tilde{z}_k = \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$

Build states manually

$$z_2 = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$z_3 = \begin{bmatrix} x_3 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$z_4 = \begin{bmatrix} x_4 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$z_5 = \begin{bmatrix} x_5 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

We write them side-by-side:

$$[z_2 \quad z_3 \quad z_4 \quad z_5] = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

THIS IS EXACTLY THE HANKEL MATRIX (SHIFTED FORM)

Earlier we wrote:

$$H = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix}$$

❗ **Why slightly different?**

Because indexing can start differently, but concept is same:

👉 Each column = **time-shifted data**

HOW TO GET THIS USING DATA (HANKEL IDEA)

We only know:

$$x = [1, 2, 3, 4, 5]$$

📌 **Step-1: Build Hankel matrix**

$$H = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix}$$

📌 **Step-2: Construct:**

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$$

$$X' = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Step 3: Compute $X^T X$

$$X^T = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 8 & 11 \\ 8 & 13 & 18 \\ 11 & 18 & 25 \end{bmatrix}$$

DMD Assumption

We assume

$$X' = AX$$

This means the matrix Amaps each snapshot to the next snapshot.

📌 **Compute**

$$A = X'X^{\dagger}$$

👉 This gives **best linear approximation**

$$X^{\dagger} = \left((X^T X)^{-1} X^T\right)$$

Step 2: Compute pseudoinverse

$$X^{\dagger} = (X^T X)^{-1} X^T$$

Step 3: Compute A

$$A = X'X^{\dagger}$$

HOW TO GET THIS USING DATA (HANKEL IDEA)

We only know:

$$x = [1, 2, 3, 4, 5]$$

📌 **Step-1: Build Hankel matrix**

$$H = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix}$$

📌 **Step-2: Construct:**

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$$

$$X' = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

❗ **IMPORTANT ISSUE**

$$X^T X \ is \ 3 \times 3$$

👉 But **rank of X = 2**

👉 So $X^T X$ is **singular (not invertible)**

📌 **Compute**

$$A = X' X^{\dagger}$$

👉 This gives **best linear approximation**

$$X^{\dagger} = \left((X^T X)^{-1} X^T \right)$$

Step 3: Compute $X^T X$

$$X^T = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 8 & 11 \\ 8 & 13 & 18 \\ 11 & 18 & 25 \end{bmatrix}$$

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ALTERNATIVE PSEUDOINVERSE FORMULA

Since X is **2 × 3**, use:

$$X^{\dagger} = X^{T}(XX^{T})^{-1}$$

STEP : COMPUTE XX^{T}

$$XX^{T} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$

Result

$$XX^{T} = \begin{bmatrix} 14 & 20 \\ 20 & 29 \end{bmatrix}$$

FIND INVERSE OF XX^{T}

$$\det = 14 \cdot 29 - 20 \cdot 20$$
$$= 406 - 400 = 6$$

$$(XX^{T})^{-1} = \frac{1}{6}\begin{bmatrix} 29 & -20 \\ -20 & 14 \end{bmatrix}$$

STEP 4: COMPUTE PSEUDOINVERSE

$$X^{\dagger} = X^{T}(XX^{T})^{-1}$$

$$X^{\dagger} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \cdot \frac{1}{6}\begin{bmatrix} 29 & -20 \\ -20 & 14 \end{bmatrix}$$

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COMPUTE $A = X'X^{\dagger}$

Given:

$$X' = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Multiply:

$$A = X'X^{\dagger}$$

First row of A

$$[2\ 3\ 4] \begin{bmatrix} -11/6 & 8/6 \\ -2/6 & 2/6 \\ 7/6 & -4/6 \end{bmatrix}$$

FINAL RESULT

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

🔴 **IMPORTANT OBSERVATION**

👉 This matrix corresponds to system:

$$x_{k+1} = 2x_k - x_{k-1}$$

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👉 ***"Hankel DMD converts time-series data into state vectors and allows us to recover the hidden system matrix using pseudoinverse."***